CE

CO2 Laser Power Supply

User Manual & Safety Instructions

Doc: EU-2024-CO2-PSU-V1.0	**Date:** March 2024	**Page:** 1/15

Product Overview

This manual covers CO2 laser power supplies manufactured by **Cloudray (Jiangsu) Co., Ltd.** These power supplies are designed for industrial CO2 laser cutting and engraving systems, providing stable high-voltage output to drive CO2 laser tubes with power ratings from 40W to 180W.

Product Range

MYJG Series

40W - 50W

Entry-level power supply

DY Series

80W - 150W

Professional grade

M Series

40W - 180W

High-performance

T Series

100W

Premium grade

⚠️ WARNING / SAFETY INFORMATION

⚠️ CRITICAL SAFETY WARNINGS

Failure to follow these warnings may result in serious injury or death.

⚠️ **WARNING: Risk of electric shock. High voltage!**

This product contains hazardous voltages. Disconnect all power sources before installation, maintenance, or troubleshooting. Allow at least 5 minutes for internal capacitors to discharge.

⚠️ **WARNING: Qualified personnel only!**

Only qualified personnel should install or service this power supply. Improper installation may result in fire, electric shock, or serious injury.

⚠️ **WARNING: Ensure proper grounding!**

This power supply must be connected to a grounded outlet. Do not remove or defeat the ground pin.

⚠️ **WARNING: Avoid exposure to water or moisture!**

Avoid exposure to water, moisture, or conductive contaminants. Operate only in a clean, dry environment.

⚠️ **WARNING: Do not operate with cover removed!**

Do not operate with the cover removed. Live parts are exposed when the cover is open.

⚠️ **WARNING: Fire hazard!**

This power supply generates high voltages that can ignite flammable materials. Keep away from combustible substances.

⚠️ **WARNING: Keep out of reach of children!**

This product is not a toy. Keep out of reach of children.

Applicable Regulations and Harmonized Standards

This product complies with the following European Union directives and regulations:

Regulation/Directive	Reference Number	Scope
General Product Safety Directive	2001/95/EC	General Product Safety

Regulation/Directive	Reference Number	Scope
Machinery Directive	2006/42/EC	As amended by (EU) 2023/1230
Toy Safety Directive	2009/48/EC	Where applicable
Low Voltage Directive	2014/35/EU	Electrical Equipment Safety
EMC Directive	2014/30/EU	Electromagnetic Compatibility
Consumer Rights Directive	2011/83/EU	Consumer Protection
Digital Services Act	DSA Regulation	Online Platform Compliance
PPE Regulation	(EU) 2016/425	Personal Protective Equipment
RED Directive	2014/53/EU	Radio Equipment (if applicable)
RoHS Directive	2011/65/EU	Hazardous Substances Restriction
REACH Regulation	(EC) No 1907/2006	Chemicals Registration
WEEE Directive	2012/19/EU	Waste Electrical Equipment

Manufacturer Declaration: This product bears the CE marking and complies with all applicable EU directives listed above. This declaration is issued under the sole responsibility of the manufacturer.

Product Specifications - All Models

The following tables provide detailed specifications for all CO2 laser power supply models:

MYJG Series (Entry-Level)

Parameter	MYJG-40W	MYJG-50W
Input Voltage	AC 110V/220V (Selectable)	AC 110V/220V (Selectable)
AC Frequency	47 ~ 440Hz	47 ~ 440Hz
Max Output Voltage	20kV	24kV
Max Output Current	20mA	20mA
Dimensions (L x W x H)	165 x 142 x 60 mm	165 x 142 x 60 mm
Weight	1.2 kg	1.3 kg
Efficiency	≥ 91%	≥ 91%
MTBF	≥ 30,000 hours	≥ 30,000 hours
Response Speed	≤ 1ms	≤ 1ms

Operating Temperature	-30°C ~ +65°C	-30°C ~ +65°C
Relative Humidity	20 ~ 85%RH (non-condensing)	20 ~ 85%RH (non-condensing)
Cooling Method	Fan cooling	Fan cooling
High Temperature Test	Full Load / 60°C / 12 hours	Full Load / 60°C / 12 hours
Start/Stop Test	500 times 7 seconds	500 times 7 seconds
Protection	Overcurrent, Open-circuit	Overcurrent, Open-circuit

DY Series (Professional Grade)

Parameter	DY-10 (80W)	DY-13 (100W)	DY-20 (150W)
Input Voltage	AC 115V/230V	AC 115V/230V	AC 115V/230V
AC Frequency	47 ~ 440Hz	47 ~ 440Hz	47 ~ 440Hz
Max Output Voltage	35kV	40kV	50kV
Max Output Current	25mA	32mA	35mA
Dimensions (L x W x H)	198 x 159 x 84 mm	270 x 163 x 85 mm	313 x 200 x 82 mm
Weight	1.7 kg	2.0 kg	3.1 kg
Efficiency	≥ 90%	≥ 90%	≥ 90%
MTBF	≥ 10,000 hours	≥ 10,000 hours	≥ 10,000 hours
Response Speed	≤ 1ms	≤ 1ms	≤ 1ms
High-Level Voltage	≥ 3V	≥ 3V	≥ 3V
Low-Level Voltage	≤ 0.7V	≤ 0.7V	≤ 0.7V
Operating Temperature	-10°C ~ +40°C	-10°C ~ +40°C	-10°C ~ +40°C
Cooling Method	Forced air cooling	Forced air cooling	Forced air cooling
High Temperature Test	Full Load / 60°C / 12 hours	Full Load / 60°C / 12 hours	Full Load / 60°C / 12 hours
Start/Stop Test	500 times 7 seconds	500 times 7 seconds	500 times 7 seconds

M Series (High-Performance)

Parameter	M40	M50	M60	M80
Input Voltage	AC 110V/220V	AC 110V/220V	AC 115V/230V	AC 115V/230V
AC Frequency	47 ~ 440Hz	47 ~ 440Hz	47 ~ 440Hz	47 ~ 440Hz
Max Output Voltage	20kV	24kV	24kV	26kV

Max Output Current	20mA	20mA	25mA	28mA
Dimensions (L x W x H)	165 x 142 x 60 mm	165 x 142 x 60 mm	197 x 161 x 91 mm	230 x 161 x 91 mm
Weight	1.2 kg	1.3 kg	2.1 kg	2.8 kg
Efficiency	≥ 91%	≥ 91%	≥ 91%	≥ 91%
MTBF	≥ 30,000 hours	≥ 30,000 hours	≥ 30,000 hours	≥ 30,000 hours
Response Speed	≤ 1ms	≤ 1ms	≤ 1ms	≤ 1ms
Operating Temperature	-30°C ~ +65°C	-30°C ~ +65°C	-30°C ~ +65°C	-30°C ~ +65°C
Cooling Method	Fan cooling	Fan cooling	Forced air cooling	Forced air cooling
External Ammeter	No	No	Yes (Included)	Yes (Included)

Parameter	M100	M120	M150	M180
Input Voltage	AC 115V/230V	AC 115V/230V	AC 115V/230V	AC 115V/230V
AC Frequency	47 ~ 440Hz	47 ~ 440Hz	47 ~ 440Hz	47 ~ 440Hz
Max Output Voltage	28kV	28kV	34kV	34kV
Max Output Current	30mA	32mA	38mA	38mA
Dimensions (L x W x H)	230 x 161 x 91 mm	230 x 161 x 91 mm	305 x 161 x 91 mm	305 x 161 x 91 mm
Weight	2.8 kg	3.3 kg	3.6 kg	4.1 kg
Efficiency	≥ 91%	≥ 91%	≥ 91%	≥ 91%
MTBF	≥ 30,000 hours	≥ 30,000 hours	≥ 30,000 hours	≥ 30,000 hours
Response Speed	≤ 1ms	≤ 1ms	≤ 1ms	≤ 1ms
Operating Temperature	-30°C ~ +65°C	-30°C ~ +65°C	-30°C ~ +65°C	-30°C ~ +65°C
Cooling Method	Forced air cooling	Forced air cooling	Forced air cooling	Forced air cooling
External Ammeter	Yes (Included)	Yes (Included)	Yes (Included)	Yes (Included)

T Series (Premium Grade)

Parameter	T100 V2.0
Input Voltage	AC 115V/230V (Selectable)
AC Frequency	50/60Hz
Max Output Voltage	40kV
Max Output Current	32mA
Dimensions (L x W x H)	283 x 170 x 102 mm

Weight	2.9 kg
Efficiency	≥ 92%
Response Speed	≤ 1ms
High-Level Voltage	≥ 3V
Low-Level Voltage	≤ 0.7V
Operating Temperature	-30°C ~ +65°C
Cooling Method	Air-cooling (Taiwan ADDA fan, Japan NMB bearing)
High Temperature Test	Full Load / 60°C / 12 Hours
Start/Stop Test	500 times 7 seconds
Special Features	LCD Display, RJ45 Interface, Adjustable Resistance

Model Comparison Summary

Series	Power Range	Input Voltage	Max Output	Cooling	Ammeter	Application
MYJG	40W - 50W	110V/220V	20-24kV	Fan	No	Entry-level
DY	80W - 150W	115V/230V	35-50kV	Forced Air	No	Professional
M (40-80W)	40W - 80W	110V/230V	20-26kV	Fan/Forced	60W+	High-performance
M (100-180W)	100W - 180W	115V/230V	28-34kV	Forced Air	Yes	Industrial
T	100W	115V/230V	40kV	Premium Air	LCD Display	Premium

Operating Instructions

Correct Usage Methods and Operating Procedures

1. **Step 1: Inspection**

   After unpacking, inspect the power supply for any visible damage. Check that all terminals are intact and confirm that the input voltage matches your local power grid. Verify that the model number corresponds to your laser tube power rating.

2. **Step 2: Installation**

   Install the power supply securely in a well-ventilated, dust-free, vibration-free cabinet. Connect the input power, ground wire, control signals, and laser tube high-voltage output cables according to the wiring diagram. Ensure all connections are tight and secure.

3 Step 3: Pre-Power Check

Reconfirm all wiring is correct, especially ensuring the high-voltage output terminals are well insulated with no exposed conductors. Verify that the cooling system (water cooling) is properly connected and operational before applying power.

4 Step 4: Testing

First perform low-voltage control signal testing without load or with a dummy load to confirm normal control operation. Then gradually increase power for loaded testing, observing whether current and voltage indicators remain stable.

5 Step 5: Operation

Use only within the rated parameters of your laser equipment. Avoid prolonged full-power operation and ensure proper heat dissipation. Monitor the power supply temperature during extended use.

Usage Restrictions

❌ This power supply is designed exclusively for CO_2 laser tubes. It is strictly prohibited to use it for any other purpose.

❌ Operating Environment: Temperature 0°C ~ 40°C, Relative Humidity ≤ 80% (non-condensing).

❌ Strictly prohibited to use in flammable or explosive environments.

❌ If the power supply shows any abnormalities (smoke, unusual noise, overheating), immediately disconnect power and stop using it.

❌ Do not exceed the rated output voltage and current specifications.

❌ Do not operate with damaged cables or connectors.

❌ Do not block ventilation openings or operate without proper cooling.

❌ Do not attempt to repair or modify the power supply yourself.

Operating Limits: Exceeding specified power, temperature, or environmental limits will void warranty and may cause immediate product failure or safety hazards.

Product Features

- **Zero-current half-bridge soft-switching circuit** - High efficiency and fast response. Compatible with various manufacturers of laser tubes.
- **Simple Port Control** - TTL level signals can control laser start/stop operations.
- **Protection Switch** - Built-in protection for external water flow, ventilation, and other safety interlocks.
- **Laser Power Adjustment** - Input 0V to 5V analog signal, can also input PWM signal for power control.
- **Open Circuit Protection** - Prevents damage to power supply due to laser tube burst.
- **Factory Aging Test** - Each power supply undergoes 60°C and 12 hours aging test under full load, plus 500 times 7-second start/stop test.
- **High-Voltage Safety** - All high-voltage lines are UL certified with gold-plated terminals for stable output.
- **External Ammeter** (M60 and above) - Network port for external LCD current display to show information status and regulate current.

Troubleshooting Guide

Problem	Possible Cause	Solution
No laser output	Power supply not powered on	Check input power connection
No laser output	Water protection activated	Check water cooling system
Weak laser power	Incorrect current setting	Adjust output current
Weak laser power	Laser tube aging	Replace laser tube
Power supply overheating	Ventilation blocked	Clear ventilation openings
Power supply overheating	Fan malfunction	Check cooling fan operation

Unstable output	Loose connections	Check and tighten all terminals
Laser tube arcing	High voltage insulation damaged	Replace high voltage cable

Maintenance

Daily Inspection (Before Operation)

- Visual inspection of power supply for damage or contamination
- Verify secure mounting (no loose screws)
- Check cooling fan operation
- Verify safety interlock functionality

Monthly Maintenance

- Clean dust from ventilation openings and fan
- Inspect all cable connections for tightness
- Check high-voltage cable insulation condition
- Verify current and voltage display accuracy

Replacement Criteria

- Replace immediately if smoke, burning smell, or visible damage is observed
- Replace if output current drops more than 10% from rated value
- Replace if cooling fan fails to operate
- Replace after 5 years of continuous operation regardless of condition

Disposal & Environmental Information

⚠️ ♻️ Environmental Notice

This product complies with WEEE Directive 2012/19/EU. Do not dispose of in general waste. Please recycle at designated electronic waste collection points.

Product	Waste Code (EWC)	Disposal Method
Power Supply (damaged/expired)	20 01 36 (Electronic waste)	WEEE recycling center
Packaging Materials	20 01 39 (Plastics), 20 01 01 (Paper)	Municipal recycling

RoHS Compliance: This product complies with Directive 2011/65/EU regarding restriction of hazardous substances. Lead (Pb), Mercury (Hg), Cadmium (Cd), Hexavalent Chromium (CrVI), PBB, PBDE, and phthalates are all below 0.1% by weight (Cadmium below 0.01%).

Manufacturer Information

Cloudray (Jiangsu) Co., Ltd.

Website: cloudray.aliexpress.com

Location: Jiangsu, China

Support: Via AliExpress platform

For EU regulatory matters, contact through the manufacturer's AliExpress store or designated EU importer.

EU Declaration of Conformity

We, **Cloudray (Jiangsu) Co., Ltd.**, declare under our sole responsibility that the CO2 laser power supplies described in this manual are in conformity with the following EU directives and harmonized standards:

- Machinery Directive 2006/42/EC (as amended by (EU) 2023/1230)
- EMC Directive 2014/30/EU
- Low Voltage Directive 2014/35/EU
- General Product Safety Directive 2001/95/EC
- RoHS Directive 2011/65/EU
- REACH Regulation (EC) No 1907/2006

This product bears the CE marking.

Version: EU-2024-CO2-PSU-V1.0 **Issue Date:** March 2024 **Next Review:** March 2025